

Adult odonate abundance and community assemblage measures as indicators of stream ecological integrity: A case study

Daniel de paiva Silva^a, Paulo De Marco^{b,*}, Daniela Chaves Resende^c

^a Departamento de Biologia Animal, Instituto de Biologia, Universidade Estadual de Campinas, UNICAMP, Rua Charles Darwin s/n, Cidade Universitária Zeferino Vaz, Campinas, Caixa Postal 6109 – CEP: 13083-970, São Paulo, Brazil

^b Laboratório de Ecologia Teórica e Síntese, Departamento de Ecologia, Universidade Federal de Goiás, UFG, Rodovia Goiânia-Nerópolis, Km 5, Campus II, Setor Itatiaia, Goiânia, Caixa Postal 131 – CEP: 74001-970, Goiás, Brazil

^c Laboratório de Bioinformática e Evolução, Departamento de Biologia Geral, Universidade Federal de Viçosa, UFV, Av. P.H. Rolfs, s/n, Viçosa, Minas Gerais, CEP: 36.570-000, Brazil

ARTICLE INFO

Article history:

Received 2 July 2009

Received in revised form 30 November 2009

Accepted 14 December 2009

Keywords:

Community indicators

Urban environment

Community structure

Species richness

Ecological integrity

ABSTRACT

Water resources demand constant conservation actions due to several problems (e.g. riparian vegetation cut-off, construction of dams, acidification, sewage and pesticide spills) that degrade the aquatic systems worldwide and affect its physicochemical parameters and habitat characteristics. Odonata is a potential group of organisms that could indicate these habitat alterations once they have aquatic and terrestrial life forms. In this study, we tested the use of adult odonate individual species and community assemblage measures to evaluate the effect of riparian vegetation cut-off and sewage discharges. The study was performed at Turvo Sujo River, in Viçosa, Southern Brazil. We selected twelve sites, six of them were upstream and six were downstream the city. Species abundance and species richness estimates of adult odonates were performed on sunny days during summer and winter. We analyzed the goodness-of-fit of the species abundances to geometric and lognormal series. We also measured the Habitat Physical Integrity Score (HPIS), pasture and forest proportions and physicochemical water parameters at each site. Only few species were abundant in up- and downstream regions. Abundance of *Argia modesta* was higher at the upstream ($t = 3.188$; $df = 17$; $p = 0.005$) than at the downstream region and this species is a potential habitat bioindicator organisms. Species richness was statistically different only in the wet season and species–abundance relations at the two regions fitted well to both geometric and lognormal series. The lack of riparian vegetation indicates a loss of habitat integrity and heterogeneity at Turvo Sujo River basin, which was mainly dominated by lake-dwelling odonate species. Low species richness differences are caused by species pool biases toward those ones capable to survive at degraded ecosystems, suggesting that the effects of water parameters are much less important than a landscape dominated by pastures and practically without forests. We suggest the use of species–abundance models (like geometric and lognormal series) to determine the degree of impacts over a given community once they are simple models and can show intrinsic processes structuring communities.

© 2009 Elsevier Ltd. All rights reserved.

1. Introduction

In mostly of terrestrial surface, the conversion of native vegetation into areas of different economic uses produced human-dominated landscapes with different impact degrees (e.g. Bennett, 1990; Blair, 1996; Mottet et al., 2006). Especially near urban areas, human impacts (e.g. higher inputs of pollutants, construction of dams, eutrophication processes and riparian vegetation cut-off) are frequently causing degradation processes that decrease the water resources' value for both present and future uses (Caspers and Heckman, 1982; Watson et al., 1982;

Delacerda et al., 1991; Bhardwaj and Tyagi, 1993; Kalcounis-Rueppell et al., 2007). All these impacts affect water quality, turning it inappropriate for human consumption and causing severe economic costs, including the search for new sources that are usually farther and more expensive.

The concept of biological integrity is a tool that can help to improve the efficiency of water management programs, since it deals with the idea that an ecosystem is able to maintain its properties even under impact (Karr, 1991). Environmental integrity is linked to the idea that, even after a disturbance, a biological system could maintain its integrity, since it preserves the majority of its components (species) and functions (interactions among these species) (Kay, 1993; Angermeier and Karr, 1994).

Aquatic insects are usually used as bioindicators, especially under an ecological integrity framework (Arimoro and Ikomi,

* Corresponding author. Tel.: +55 62 3521 1480.

E-mail addresses: pdemarco@icb.ufg.br, pdemarco@pq.cnpq.br (P. De Marco).

2009; Blocksom and Johnson, 2009; Leunda et al., 2009). A recent study had showed a variety of macroinvertebrate groups that could be used as reliable indicators of watershed urbanization (Utz et al., 2009). Nevertheless, those studies use samples of those indicators restricted to its aquatic environments. Some of these aquatic groups (e.g. Odonata and Coleoptera) are also present in the terrestrial areas near the streams, usually as adults. For instance, once adult odonates defend mating territories near water, they are strongly selected to choose habitats where its larvae (Wolf and Waltz, 1988; Alcock, 1990) may have higher survival and, thus, this group may be an excellent indicator for modifications occurring both in the aquatic and the terrestrial environment.

Odonata have aquatic larval instars with strong dependence on the substratum type, such as the aquatic vegetation present at the water bodies (Corbet, 1980, 1999). The larvae usually live buried under the sediments or hidden on the aquatic vegetation, increasing both the chances to obtain food and protection against its predators (Westfall, 1984; Johansson, 1991). Adult males of territorial odonate species also maintain dependence on the aquatic environment, defending areas where females are easier to find (Jacobs, 1955; Conrad and Pritchard, 1992). Adult females visit water bodies to copulate and to oviposit (Michiels and Dhondt, 1989, 1991), selecting sites based on the resources quality (Alcock, 1987) that warrant a higher larvae survival (Harvey and Corbet, 1985). This mating system based on territorial defense and disputes for females, generates thermoregulatory restrictions to these organisms (May, 1976, 1991; De Marco, 1998; De Marco and Resende, 2002) in a way that physical characteristics of the water bodies (i.e. presence of riparian vegetation, quantity of light and stream or river width) could affect their species composition.

Carle (1979) considers that Odonata species are good “thermometers” of the environment quality because: (1) they inhabit all freshwater aquatic habitats, (2) their larvae have specific tolerance to environmental disturbances, (3) many species had a larval stage which lasts at least one year, (4) the larvae are relatively sedentary, (5) the dispersal of adult individuals is fast, so they can rapidly reestablish populations on adequate environments and, finally, (6) the adults can be easily observed due to their territorial behavior on water bodies and, finally, (7) the larvae and the adults can be easily identified. Actually, this last item is not completely true for neotropical odonates since the majority of larvae are not yet properly described. Additionally, Oertli (2008) includes the facility to do quantitative studies and its ability to generate political sympathy as important issues of odonates as bioindicators. Considering all these properties, the odonate assemblage may be considered as a surrogate for the insect community structure near water bodies, being capable to indicate changes in the biological integrity of these systems.

Proximity to urban areas may cause loss of biological integrity of streams both through water parameters change (Watson et al., 1982; Muttamara, 1996; Cloern, 1999; Smith et al., 1999; Mainstone and Parr, 2002) and landscape conversion from forest to pastures caused by economic pressures (Biggs et al., 2008; Casatti et al., 2009). No matter what causes the environmental alterations, the biological integrity may be altered in a variety of ways including the loss of species diversity (Paoletti, 1999; Hillebrand and Blenckner, 2002), the increase of the population size of some pest species (Kogan and Lattin, 1993; Naranjo et al., 2004) or the decrease of habitat quality (Collinge, 1996; Summerville and Crist, 2001; Steffan-Dewenter and Tscharnkte, 2002; Utz et al., 2009). In a higher scale, species richness and abundance relations may respond to alterations in a predictive way with loss of biodiversity and increased dominance of tolerant species (May, 1982; Death, 1996; Hill and Hamer, 1998).

Current ecological knowledge of larval taxonomy may be at the most optimistic guess nearly 20–30% of the actual neotropical

species richness (De Marco and Darc, unpublished results). The same general problem affects the study of odonate indicator properties for south-African dragonflies (Smith et al., 2007). Under this constraint, we consider that the odonate adult assemblage must be a surrogate for the patterns ongoing with the larvae assemblage. We based our argument in two assumptions. Firstly, the larvae present in a given site are directly related to the adults that defend territories and lay eggs in this site, at least in headwater streams that are not affected by larval import through drift. Secondly, the group of organisms selected as ecological indicators should be efficient, with well known relations between them and the properties of the ecosystem which they are expected to indicate. Ecological information for tropical species is concentrate on adults, with many recent studies dealing with general thermoregulatory patterns (May, 1977; De Marco et al., 2005; Corbet and May, 2008) and some studies about its ecological indicator properties in this system (Clark and Samways, 1996; Ferreira-Peruquetti and De Marco, 2002; Suhling et al., 2006; Smith et al., 2007; Corbi et al., 2008).

In this paper we consider the effects of urban areas both on (1) general landscape changes (related to the loss of original forest and its conversion to pasture) and on (2) water parameters alterations due to sewage load. Therefore, we considered two stream regions: the upstream one (before urban area) and the downstream one (after urban area). The proximity of the water bodies to the cities increases the chance of impacts as riparian vegetation cut-off and soil run-off. As these impacts tend to be higher near the city, the downstream environment would become different from the upstream one, causing changes on the characteristics of the Odonata assemblages. The discharges of sewage and other pollutants on the water bodies can alter water physicochemical parameters (e.g. conductivity, dissolved oxygen, turbidity and pH), especially in the downstream region (Watson et al., 1982; Simiao-Ferreira et al., 2009).

We evaluate the use of some individual species and some measures of community structure of adult odonate to indicate urban environmental impacts effects at the Turvo Sujo basin at Viçosa, Southeastern Brazil. Specifically, we tested if those measures could indicate different alterations as (i) physical integrity of nearby stream banks; (ii) water parameters variables; (iii) land-use variables; (iv) individual species abundance; (v) species richness and (vi) species–abundance relation of the odonate assemblages.

2. Study sites

We studied Turvo Sujo River, a water body that pass through Viçosa, state of Minas Gerais, Southeastern Brazil (20°45'14"S, 45°52'54"W). Both stream regions, upstream and downstream of the city, presented an advanced stage of environmental degradation with loss of riparian vegetation at the river margins and conversion of the native Atlantic Forest to pastures, what facilitates the soil run-off processes. The smell of water was unpleasant at some sites, indicating the occurrence of eutrophication processes.

Following the Köppen classification, the region weather was classified as Cwa, humid mesothermic, with hot summers and dry winters. The mean temperature of the hottest month is higher than 22 °C, while in the coldest one it is around 10 °C. The mean annual rainfall is between 1500 and 2000 mm, with relative moisture around 80% (Valverde, 1958). The rainy season occurs from October until March with peaks on November and January (Golfari, 1975).

3. Methods

We chose the study sites from a satellite image to assure that the distance between consecutive study sites at the river was, at

least, 500 m. The twelve sites studied at Turvo Sujo River consisted of linear transects of 100 m each, six of them were located upstream while the other six were located downstream the city. The downstream sites were nearer the city in order to avoid the interference of river's self-cleaning processes on the sewage effects over the adult odonates assemblages.

Each site was divided into 20 segments, with 5 m length, where we analyzed the adult odonate faunistic composition, by visual scans. We estimated the abundance of almost all species counting the number of individuals in each segment instantaneously. The

individuals of species undistinguishable at the field (especially *Argia* and some Protoneuridae) were captured and identified at the laboratory using available taxonomic keys and species descriptions (May, 1998; Carvalho and Calil, 2000; Costa et al., 2002; Lencioni, 2005, 2006). The scans were realized on sunny days, between 1200 and 1500 h, in two different days for each site. Before the study, field researchers were trained to recognize the species present in the field, assuring a rapid identification and the efficiency in abundance estimations. These methods have been successfully used in many ecological and behavioral studies of Odonata (De

Table 1
Habitat Physical Integrity Score (HPIS).

Factor	Characteristic	Condition	Score
F1	Land use pattern beyond the Riparian Zone	Primary continue forest/100 ha fragment/10 ha fragment	1
		Secondary forest	2
		Pasture	3
		Perennial crops	4
		Short-cycle crops/exposed soil	5
F2	Width of riparian forest	Continuous forest	1
		Forest width between 30 and 100 m	2
		Forest width between 5 and 30 m	3
		Forest width between 1 and 5 m	4
		Riparian forest absent, but some shrub species and pioneer trees	5
		Riparian forest and shrub vegetation absent	6
F3	Completeness of riparian forest	Riparian forest intact without breaks in vegetation	1
		Breaks occurring at intervals of >50 m	2
		Breaks frequent with gullies and scars at every 50 m	3
		Deeply scarred with gullies all along its length	4
F4	Vegetation of riparian zone within 10 m of channel	More than 90% plant density by non-pioneer trees or shrubs	1
		Mixed pioneer species and mature trees	2
		Mixed grasses and sparse pioneer trees and shrubs	3
		Grasses and few tree shrubs	4
F5	Retention devices	Channel with rocks and/or old logs firmly set in place	1
		Rocks and/or logs present but backfilled with sediment	2
		Retention devices loose, moving with floods	3
		Channel of loose sandy silt, few channel obstructions	4
F6	Channel sediments	Little or no channel enlargement resulting from sediment accumulation	1
		Some gravel bars of coarse stones and little silt	2
		Sediment bars of rocks, sand and silt common	3
		Channel divided into braids or stream channel corrected	4
F7	Bank structure	Banks inconspicuous	1
		Banks stable, with rock/soil held firmly by grasses, shrubs or tree roots	2
		Banks firm but loosely held by grasses and shrubs	3
		Banks of loose soil held by a sparse layer of grass and shrubs	4
		Banks unstable, easily disturbed, with loose soil or sand	5
F8	Bank undercutting	Little, not evident or restricted to areas with tree root support	1
		Cutting only on curves and at constrictions	2
		Cutting frequent, undercutting of banks and roots	3
		Severe cutting along channel, banks falling in	4
F9	Stream bottom	Stone bottom of several sizes packed together, interstices obvious	1
		Stone bottom easily moved, with little silt	2
		Bottom of silt, gravel and sand, stable in some places	3
		Uniform bottom of sand and silt loosely held together	4
		Stony substrate absent	5
F10	Riffles and pools, or meanders	Distinct, occurring at intervals of 5–7x the stream width	1
		Irregularly spaced	2
		Long pools separating short riffles, meanders absent	3
		Meanders and riffle/pools absent or stream corrected	4
F11	Aquatic vegetation	When present, consists of moss and patches of algae	1
		Algae dominant in pools, vascular plants along edge	2
		Algal mats present, some vascular plants, few mosses	3
		Algal mats cover bottom, vascular plants dominate channel	4
		Aquatic vegetation absent	5
F12	Detritus	Mainly consisting of leaves and wood, without sediment	1
		Mainly consisting of leaves and wood, with sediment	2
		Few leaves and wood, fine organic debris, without sediment	3
		No leaves or woody debris, coarse/fine organic matter, with sediment	4
		Fine anaerobic sediment, no coarse debris	5

Marco, 1998; Ferreira-Peruquetti and De Marco, 2002; Ferreira-Peruquetti and Fonseca-Gessner, 2003).

There is no important stream system near Turvo Sujo River that may function as source of adults for the area. We also observed the presence of exuvia in those streams regularly in qualitative observations in the area over the past 10 years. Previous studies also reported persistent populations of some stream-dweller species as *Hetaerina rosea* and *Tigriagrion aurantinigrum* in some of the studied sites (Ferreira-Peruquetti and De Marco, 2002; De Marco and Peixoto, 2004; De Marco and Vital, 2008). These observations suggest that many of the adults in the area may develop from larvae from the same site or nearby sites of the same stream system.

Analyses of the physical characteristics of the sites were carried out according to a Habitat Physical Integrity Score (HPIS) (Table 1, modified from Nessimian et al., 2008) that includes characteristics of the riparian forest, river banks and channel structure that might be related to human impacts. A score for each element is assigned according to Table 1 and the score is calculated as the sum of these values. The score produced is inversely related to the physical integrity of the site. Sixty samples of water current were measured at each region, upstream and downstream the city of Viçosa using an AA Current meter Scientific Instruments, Inc. We performed chemical analyses of the water parameters such as conductivity, turbidity, dissolved oxygen and pH on the dry season from five samples arbitrarily chosen at each 100 m transect. These samples were analyzed at the Water Treatment Station of Universidade Federal de Viçosa, less than 3 h after collection.

We derived the land cover around the sites using a maximum likelihood classification of an Aster satellite image of the area with 15 m pixel size. The categories of the classification were forest, pasture, coffee plantation, *Eucalyptus* plantation, bare soil and urban area. In general, these categories are easily discriminated on the image, except by the confusion between the forest border, *Eucalyptus* and coffee plantations. Landscape analysis including proportion of those categories around the sites were done using ArcView 3.2 (ESRI, 1999).

3.1. Statistical analyses

In all tests a 5% significance level was used. For comparisons between downstream and upstream areas we used independent *t*-tests, and when the homogeneity of variances assumption was not supported, we used independent *t*-test with separate variances estimates (Zar, 1999). We used standard linear regression procedures (Zar, 1999) to test the relationships between some biotic variables (such as species richness and abundance) with the land cover and the HPIS.

We estimated the species richness using a Jackknife procedure to correct the intrinsic bias present on this measure (Heltshe and Forrester, 1983; Coddington et al., 1991; Colwell and Coddington, 1994). Jackknife is a re-sampling technique that provides an

estimate for the species richness and for its 95% confidence interval and was estimated using the ESTIMATE S software (Colwell, 2000).

Species abundance patterns were determined by the fit to the geometric (May, 1975) and lognormal distributions (Preston, 1948; Preston, 1962). We used nonlinear estimation methods based on quasi-newton interactive procedure and on least squares loss function. The fit to these distributions was evaluated using standard Chi-square goodness-of-fit tests.

The geometric series have only a parameter *k* that represent the proportion of resource allocated to each species and is described by the following equation (according to Magurran, 1988):

$$n_i = NC_k k(1 - k)^{i-1} \quad (1)$$

where n_i = the number of individuals of the species that occupies the *i*th abundance position; *N* = total number of individuals; $C_k = [1 - (1 - k)^s]^{-1}$ is a constant that assures that $\sum n_i = N$.

The lognormal distribution is described by

$$S(R) = S_0 \exp(-a^2 R^2) \quad (2)$$

where $S(R)$ = number of species on the *r*th octave; S_0 = number of species on the modal octave; $a = (2\sigma^2)^{1/2}$ = inverse amplitude of the distribution.

4. Results

4.1. Environmental variation related to urban presence

The stream did not present aquatic and riparian vegetation at the study sites, lacking shaded areas that are frequently used as perches for many males and females odonate near the water. HPIS revealed a high level of physical degradation, but there were no upstream/downstream statistical differences (Table 2). In general, the sites presented a high proportion of pastures (>0.70) and a low proportion of forest, indicating landscape degradation. Pasture proportion is higher in the sites sampled downstream, but there is no change on the proportion of forest around the city (Table 2).

The water parameters were statistically equal according to *t*-test, except for dissolved oxygen, that presented a higher concentration in the upstream region (Table 2). pH presented a higher variability upstream and conductivity a higher variability downstream (statistically significant Levene test).

4.2. General patterns of species assemblage composition

We observed 28 taxa of Odonata belonging to Calopterygidae, Coenagrionidae and Protoneuridae (from Zygoptera suborder), and to Gomphidae and Libellulidae (from Anisoptera suborder) families, in the 12 studied sites. All of these taxa presented an abundance standard deviation higher than the mean, indicating an aggregated distribution in the segments (Table 3).

Table 2

Physicochemical parameters, biotic integrity analyses and landscape analysis performed during the dry season at Turvo Sujo River, Viçosa, Southeastern Brazil. SD = standard deviation.

Variable	Upstream		Downstream		t-Test		
	Mean	SD	Mean	SD	<i>t</i>	df	<i>p</i>
pH	6.958	0.155	6.855	0.032	1.584	6	0.169
OD (g l ⁻¹)	8.404	0.284	5.597	0.517	8.569	7	<0.010
Conductivity (μS cm ⁻¹ , 25 °C)	56.176	1.161	62.860	16.269	-1.003	6	0.361
Turbidity (NTU)	17.996	3.232	22.836	17.684	-0.659	10	0.524
Water current (m s ⁻¹)	19.166	7.612	26.333	6.463	-1.757	10	0.109
Habitat Physical Integrity Score (HPIS)	43.666	1.505	44.333	0.816	-0.953	10	0.362
Forest proportion	0.070	0.143	0.022	0.035	1.117	12	0.285
Pasture proportion	0.690	0.209	0.877	0.086	-2.847	15	0.012

Table 3

Mean abundance of the species present at the sites throughout the Turvo Sujo River, Viçosa Southeastern Brazil.

Taxa		Mean	Standard Deviation
Coenagrionidae			
<i>Argia modesta</i>	Calvert, 1909	0.343	0.737
<i>Tigriagrion aurantigrum</i>	Calvert, 1909	0.045	0.219
<i>Acanthagrion ascendens</i>	Calvert, 1909	0.035	0.225
<i>Acanthagrion gracile</i>	Rambur, 1842	0.029	0.168
<i>Acanthagrion lancea</i>	Selys, 1876	0.020	0.192
<i>Homeoura chelifera</i>	Selys, 1876	0.004	0.064
<i>Homeoura lindneri</i>	Ris, 1928	0.002	0.045
<i>Ischnura capreolus</i>	Hagen, 1861	0.002	0.045
<i>Oxyagrion</i> spp.		0.002	0.045
Calopterygidae			
<i>Hetaerina rosea</i>	Selys, 1853	1.268	1.557
Protoneuridae			
<i>Epipleoneura venezuelensis</i>	Rácenis, 1955	0.010	0.120
<i>Neoneura sylvatica</i>	Hagen in Selys, 1886	0.004	0.064
<i>Peristicta aeneoviridis</i>	Calvert, 1909	0.002	0.045
Libellulidae			
<i>Erythrodiplex fusca</i>	Rambur, 1842	0.041	0.238
<i>Elasmothermis cannaoides</i>	Calvert, 1906	0.035	0.216
<i>Macrothemis absimilis</i>	Costa, 1991	0.033	0.230
<i>Perithemis mooma</i>	Kirby, 1889	0.014	0.136
<i>Macrothemis</i> sp.		0.010	0.120
<i>Elasmothermis constricta</i>	Calvert, 1898	0.006	0.101
<i>Orthemis discolor</i>	Burmeister, 1839	0.006	0.078
<i>Pantala flavescens</i>	Hagen, 1861	0.004	0.064
<i>Dythemis multipunctata</i>	Kirby, 1894	0.002	0.045
<i>Micrathyrina hesperis</i>	Ris, 1911	0.002	0.045
<i>Micrathyrina hypodidyma</i>	Calvert, 1906	0.002	0.045
Gomphidae			
<i>Phyllocycla</i> sp.		0.002	0.045

The most abundant species throughout the River's course were *H. rosea* (1.268 ± 1.557 , mean $\pm$ standard deviation) and *Argia modesta* (0.343 ± 0.737) (Table 3). The most abundant species upstream the city were *H. rosea* (1.491 ± 1.868), *A. modesta* (0.566 ± 0.916), *Erythrodiplex fusca* (0.070 ± 0.315), *Acanthagrion ascendens* (0.037 ± 0.211), *Acanthagrion gracile* (0.037 ± 0.190) and *Elasmothermis cannaoides* (0.037 ± 0.190). In the downstream region, the most abundant species were *H. rosea* (1.045 ± 1.128), *A. modesta* (0.120 ± 0.385), *T. aurantigrum* (0.054 ± 0.226), *Macrothemis absimilis* (0.050 ± 0.284), *A. ascendens* (0.033 ± 0.239) and *E. cannaoides* (0.033 ± 0.239).

Dythemis multipunctata, *Elasmothermis constricta*, *Epipleoneura venezuelensis*, *Ischnura capreolus*, *Micrathyrina hypodidyma*, *Neoneura sylvatica*, *Oxyagrion* spp. and *Pantala flavescens* occurred only upstream the city, while *Homeoura chelifera*, *Homeoura lindneri*, *Micrathyrina hesperis*, *Peristicta aeneoviridis* and *Phyllocycla* sp. occurred only downstream the city.

At all study sites we observed both stream-dwelling species (i.e. *H. rosea*, *A. modesta*, *M. absimilis*, *T. aurantigrum*, *E. cannaoides*, *E. constricta*, Gomphidae sp., *P. aeneoviridis*) as much as lake-dwelling species (i.e. *Acanthagrion lancea*, *A. ascendens*, *A. gracile*, *Perithemis mooma*, *I. capreolus*, *M. hypodidyma*, *M. hesperis*, *Orthemis* sp.). The presence of lake-dwellers species indicates variations on water current and presence of pools throughout the river's course.

Table 4

Comparison of the abundance estimates of the species present at Turvo Sujo River, Viçosa, Southeastern Brazil. SD = standard deviation. The analysis was done pooling wet and dry season, with $n = 12$ to both upstream and downstream sites.

	Upstream		Downstream		t-Test			Variance comparison	
	Mean	SD	Mean	SD	t	df	p	F-Ratio	p
<i>A. modesta</i>	11.167	8.354	2.417	4.542	3.188	17	0.005	3.383	0.055
<i>H. rosea</i>	29.750	21.980	20.917	11.751	1.228	17	0.236	3.499	0.049

4.3. Up and downstream differences in abundances

We used only the most abundant species in the whole river course (*A. modesta* and *H. rosea*) to perform a general comparison between the upstream and downstream regions. Abundance differences can be caused by local microhabitat differences (measured by the HPIS) or by regional effects (measured by the proportion of pastures around the study sites). We did not find relationships between *H. rosea* abundance and the habitat physical integrity, in none seasons or regions studied (upstream/wet season $r^2 = 0.001$; $p = 0.947$, downstream/wet season $r^2 = 0.042$; $p = 0.696$, upstream/dry season $r^2 = 0.139$; $p = 0.465$, downstream/dry season $r^2 = 0.361$; $p = 0.206$). Similarly, this species did not respond to differences of pasture proportion in the landscape (upstream/wet season $r^2 = 0.158$; $p = 0.433$, downstream/wet season $r^2 = 0.55$; $p = 0.09$, upstream/dry season $r^2 = 0.204$; $p = 0.367$, downstream/dry season $r^2 = 0.255$; $p = 0.306$). *A. modesta*, similarly to *H. rosea*, did not present any relationships between its abundance and the habitat physical integrity (upstream/wet season $r^2 = 0.535$; $p = 0.097$, downstream/wet season $r^2 = 0.032$; $p = 0.73$, upstream/dry season $r^2 = 0.292$; $p = 0.267$, downstream/dry season $r^2 = 0.113$; $p = 0.512$) or the pasture proportion in the landscape analysis (upstream/wet season $r^2 = 0.015$; $p = 0.812$, downstream/wet season $r^2 = 0.009$; $p = 0.86$, upstream/dry season $r^2 = 0.075$; $p = 0.6$, downstream/dry season $r^2 = 0.113$; $p = 0.276$).

While *A. modesta* presented abundance four times higher at the upstream when compared to downstream region, the abundance of *H. rosea* was not different in these regions (Table 4). In both cases, the variances were higher at the upstream region (Table 4), suggesting a higher environmental heterogeneity at this region.

4.4. Up and downstream differences in species richness

A Jackknife estimate, based on all samples of the Turvo Sujo River, predicted 40 species in Turvo Sujo River course (confidence interval of ± 6.15 species). The species richness estimate in the wet season was higher at the upstream than the downstream region, but we did not observed differences between them at the dry season. The richness estimate of the wet and the dry seasons is statistically different based on inferences of the confidence interval. On the wet season, the species richness estimate was almost the double when compared with the estimate on the dry one (Fig. 1).

There is no relationship between the pasture proportion and the species richness estimate at the upstream and downstream regions in both seasons (upstream/wet season $r^2 = 0.076$; $p = 0.598$, downstream/wet season $r^2 = 0.458$; $p = 0.14$, upstream/dry season $r^2 = 0.009$; $p = 0.853$, downstream/dry season $r^2 = 0.152$; $p = 0.444$).

4.5. Species–abundance relationships

The species–abundance distribution on both regions, upstream and downstream the city, fitted to the geometric series model (Fig. 2). According to this model, at the upstream region, the proportion of preferential niche utilized by each species was 62% of the total ($r^2 = 0.98$, $p < 0.05$). At the downstream region, the proportion of preferential niche used by each species was around 76% of the total ($r^2 = 0.98$, $p < 0.05$). We did not observe confidence

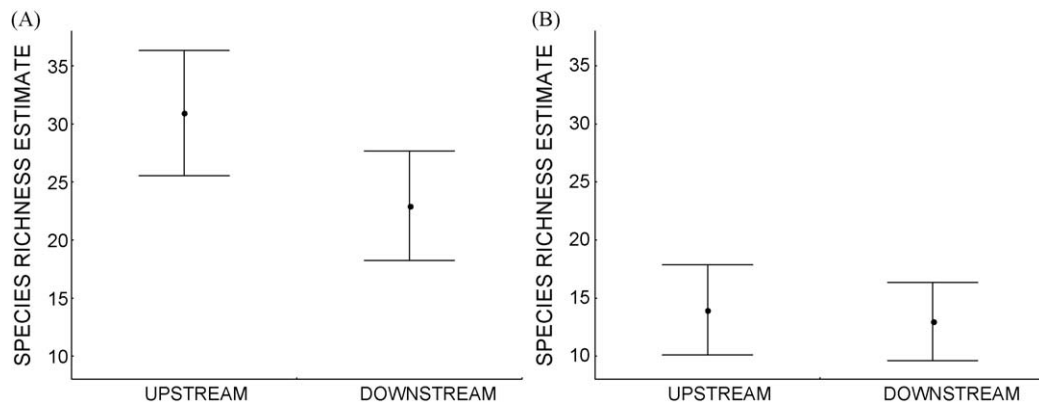


Fig. 1. Comparison of the specie richness estimate at the upstream and the downstream regions in the course of the Turvo Sujo River, Viçosa, Southeastern Brazil, during (A) the wet and (B) the dry season, using a 95% confidence interval. The analysis was done pooling wet and dry season, with $l = 12$ to both upstream and downstream sites.

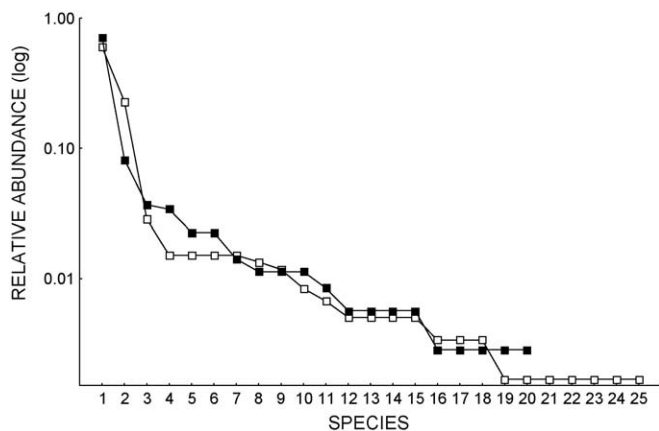


Fig. 2. Species–abundance relationship of upstream and downstream regions of Turvo Sujo River course, Viçosa, Southeastern Brazil (upstream sites: open symbols; downstream sites: closed symbols).

intervals overlap between the k estimate of both regions, indicating a higher resources monopolization by the species inhabiting the downstream region. Meanwhile, the species–abundance distribution pattern observed upstream ($So = 9.725$, standard error = 2.063; $a = 0.312$; $SE = 0.083$; $\chi^2 = 1.888$, $df = 2$, $p = 0.389$) as much as the one observed downstream ($So = 9.396$, $SE = 1.706$; $a = 0.366$; $SE = 0.075$; $\chi^2 = 0.242$, $df = 1$, $p = 0.376$) fitted to the lognormal distribution.

5. Discussion

Except for the dissolved oxygen, none of the water parameters of the urban area presented statistical significance effects. This difference in dissolved oxygen concentration may show that the wastewater input from the urban area might be accelerating the eutrophication process at the downstream sites. This may cause an increase of organic matter and microorganisms that consume dissolved oxygen, limiting the set of species capable to survive in this environment (Lachavanne et al., 1992; Pinto-Coelho, 1998). Nevertheless, all areas are considered severely impacted based on physical properties, especially by the loss of riparian vegetation.

A cautionary note must be attached to the discussion of odonate species response to water parameters variables. Many odonates have great dispersal abilities (McPeck, 1989; Anholt, 1990; Schutte et al., 1997) and could re-colonize impacted areas. Immature adults usually fly away from their natal sites (Corbet, 1999) and could go to areas where its larvae might not complete normal development (Schultz et al., 2003). Nevertheless, this could be

especially true for rare species but it is not expected to explain the patterns for the abundant *H. rosea* and *A. modesta*.

The question that arises is which impact was the most important to determine the patterns of the odonate assemblage that we have found: the landscape alterations with loss of original forest cover or the water parameters related to urbanization? Our results suggest that in a pasture dominated landscape the effect of water parameters are much less important. Foote and Hornung (2005) could not find any differences on water parameters in studied sites of a pasture dominated landscape but find that vegetation structure could be an important predictor of odonate diversity. Otherwise, the removal of forest cover may facilitate the invasion of the lake-dweller species and this appears to occur in all areas of the present study. Most streams near urban areas or in agriculture dominated landscapes are strongly modified and the forest coverage is lost. The same general results with the presence of generalist lentic-dweller species were found by Solimini et al. (1997) in a river with increasing organic pollution. They also suggest that the observed differences in assemblage composition may be explained by differential sensitivity of those species to the organic pollution.

Many odonate species are habitat-dependent and are associated to forested streams (Fincke, 1984; Taylor and Merriam, 1995; Samways and Steytler, 1996; Copeland et al., 1996; Clausnitzer, 1998; Ferreira-Peruquetti and De Marco, 2002). On the sampled fauna, those species are possibly locally extinct because Turvo Sujo River has lost its forested areas. In a region where the forest was lost for a long period of time (40–50 years in Viçosa), the species pool is biased toward those species able to survive in this system. The loss of specialized species and the abundance increase or invasion of species with less specific habitat requirements is expected in such systems (Ferreira-Peruquetti and De Marco, 2002). In species-poor regional pools, it is expected that the effects of local variables as dissolved oxygen or the physical environment (as described by HPIS) could not be observed because the species present are exactly those that are indifferent to those alterations. The same general explanation could be used to explain the inexistence of differences in the species abundance distribution in upstream/downstream comparisons. The species are indifferent to this effect and the general pattern could be described by geometric series as well as by lognormal distribution.

The main weakness of our approach is the lack of an appropriate control site with a preserved riparian forest, once the economic development in this region has converted forested areas near streams into pastures or agricultural systems. The geo-morphological characteristics of this landscape, commonly referred as “mar-de-morros” (land of hills, Valverde, 1958), constrained most of the economic uses to the more productive riparian areas, except

for coffee plantations on the hills (De Marco and Coelho, 2004). In this region, there is no proper control site due to the association of geo-morphological features and the pattern of land conversion. Therefore, our discussion is based on the expected species richness and species composition in forested streams present in the preserved State Park of Rio Doce (Ferreira-Peruquetti and De Marco, 2002). A previous comparison of the streams in Viçosa (a pasture landscape) and in the State Park of Rio Doce (a protected and forested landscape) showed the absence of many specialized odonate species as the Megapodagrionidae and a large proportion of Libellulidae species in Viçosa (Ferreira-Peruquetti and De Marco, 2002).

Some elaborate approaches are developed to the assessment of river/floodplain systems using odonate species (Oertli, 2008). For instance, the use of the Odonata Habitat Index (OHI) allow to classify the species according to its association to five general habitat types, and have been successfully used to indicate habitat alterations (Chovanec and Waringer, 2001; Schultz et al., 2003; Chovanec et al., 2004; Chovanec et al., 2005). The OHI protocol highlights the importance of temporary pools and permanent water as habitat templates for odonates in large rivers and floodplains (Schultz et al., 2003; Schindler et al., 2003). However, we expect that forested headwater streams in tropical areas maintain a rich and specific odonate assemblage, while medium to large rivers presents a higher compositional similarity to lakes or ponds (Juen et al., 2007). On headwater streams we also observed some species associations to riffles and pools which are consistent with the basic concepts of the OHI, but suggest a further development of more detailed protocols to deal of subtle habitat variations within sampled reaches that affects species composition.

Watson et al. (1982) reported strong differences caused by sewage effluents on both adult and larvae species richness and species composition. The differences on the regional species pools conditions, observed between the present study and the one performed by Watson et al. (1982), are caused by landscape degradation and by the high organic and heavy metal contamination, respectively.

Nevertheless, two important differences are observed here: the higher species richness at the upstream region and the lower adjusted k of the geometric series. These two results are consistent with the existence of a higher proportion of resources monopolized by downstream assemblages, with consequent lower species richness. Therefore, the quantity of species that this environment could stand would diminish once the resources would reach quantities lower than the minimum required to maintain the same species of the upstream region at the downstream one. The fit to a geometric series is also interpreted as a result of a single environmental factor that strongly affect community organization (e.g. Skalski and Pospiech, 2006) which is consistent with the high homogeneity of the system that resulted from the conversion from forest to pasture in this area. This general result also support the use of species–abundance models as ecological indicators of community changes (Gray, 1981; Nelson, 1987; Death, 1996; Kevan et al., 1997; Tsirtsis et al., 2008). This species distribution models are under intense theoretical debate and will be especially useful for ecological indicator analysis as long their theoretical mechanisms are clearly explained (Mcgill et al., 2007). Our analysis suggests that the use of these two models (geometric and lognormal) that may represent very different mechanisms may be a simple and powerful tool in this kind of studies.

Acknowledgements

We would like to thank Patrícia Ferreira-Peruquetti who helped in species identification and field work and Marcos Vinícius

Carneiro Vital and Luis Maurício Bini, who reviewed and early version of this manuscript. We also thank two anonymous reviewers whose contribution was very important to a previous version of this manuscript. Daniel de Paiva Silva and Paulo De Marco Jr. were granted by CNPq. Daniela Chaves Resende was granted by CAPES.

References

- Alcock, J., 1987. The effects of experimental manipulation of resources on the behavior of two calopterygid damselflies that exhibit resource-defense polygyny. *Can. J. Zool.* 65, 2475–2482.
- Alcock, J., 1990. Oviposition resources, territoriality and male reproductive tactics in the dragonfly *Palaemonetes lineatipes* (Odonata: Libellulidae). *Behaviour* 113, 251–263.
- Angermeier, P.L., Karr, J.R., 1994. Biological integrity versus biological diversity as policy directives. *BioScience* 44, 690–697.
- Anholt, B.R., 1990. Size-biased dispersal prior to breeding in a damselfly. *Oecologia* (Berlin) 83, 385–387.
- Arimoro, F.O., Ikomi, R.B., 2009. Ecological integrity of upper Warri River, Niger Delta using aquatic insects as bioindicators. *Ecol. Indic.* 9, 455–461.
- Bennett, A.F., 1990. Land use, forest fragmentation and the mammalian fauna at Naringal, southwestern Victoria (Australia). *Aust. Wildlife Res.* 17, 235–248.
- Bhardwaj, A.C., Tyagi, N., 1993. Impact of organochlorine pesticide on the morphobehaviour of *Pantala flavescens* Fabr (Libellulidae, Odonata). *J. Environ. Biol.* 14, 89–94.
- Biggs, T.W., Dunne, T., Roberts, D.A., Matricardi, E., 2008. The rate and extent of deforestation in watersheds of the southwestern Amazon Basin. *Ecol. Appl.* 18, 31–48.
- Blair, R.B., 1996. Land use avian species diversity along an urban gradient. *Ecol. Appl.* 6, 506–519.
- Blocksom, K.A., Johnson, B.R., 2009. Development of a regional macroinvertebrate index for large river bioassessment. *Ecol. Indic.* 9, 313–328.
- Carle, F.L., 1979. Environmental monitoring potential of the Odonata, with a list of rare and endangered Anisoptera of Virginia, United States. *Odonatologica* 8, 319–323.
- Carvalho, A.L., Calil, E.R., 2000. Chaves de identificação para as famílias de Odonata (Insecta) ocorrentes no Brasil, Adultos e larvas. *Papeis Avulsos de Zoologia, São Paulo* 41, 223–241.
- Casatti, L., Ferreira, C.D., Carvalho, F.R., 2009. Grass-dominated stream sites exhibit low fish species diversity and dominance by guppies: an assessment of two tropical pasture river basins. *Hydrobiologia* 632, 273–283.
- Caspers, H., Heckman, C.W., 1982. The biota of a small standing water ecosystem in the Elbe flood plain. The role of districts free from pesticide sprays as havens for endangered aquatic species. *Arch. Hydrobiol. Suppl.* 61, 227–316.
- Chovanec, A., Waringer, J., 2001. Ecological integrity of river-floodplain systems—assessment by dragonfly surveys (Insecta: Odonata). *Regul. Rivers Res. Manage.* 17, 493–507.
- Chovanec, A., Waringer, J., Raab, R., Laister, G., 2004. Lateral connectivity of a fragmented large river system: assessment on a macroscale by dragonfly surveys (Insecta: Odonata). *Aquat. Conserv. Mar. Fresh. Ecosyst.* 14, 163–178.
- Chovanec, A., Waringer, J., Straif, M., Graf, W., Reckendorfer, W., Waringer-Löschenkohl, A., 2005. The Floodplain Index: a new approach for assessing the ecological status of river/floodplain-systems according to the EU Water Framework Directive. *Large Rivers* 15, 169–185.
- Clark, T.E., Samways, M.J., 1996. Dragonflies (Odonata) as indicators of biotope quality in the Kruger National Park, South Africa. *J. Appl. Ecol.* 33, 1001–1012.
- Clausnitzer, V., 1998. Territorial behaviour of a rainforest dragonfly *Notiothemis robertsi* (Odonata: Libellulidae): proposed functions of specific behavioural patterns. *J. Zool.* 245, 121–127.
- Cloern, J.E., 1999. The relative importance of light and nutrient limitation of phytoplankton growth: a simple index of coastal ecosystem sensitivity to nutrient enrichment. *Aquat. Ecol.* 33, 3–16.
- Coddington, J.A., Griswold, C.E., Davila, D.S., Penaranda, E., Larcher, S.F., 1991. Designing and testing sampling protocols to estimate biodiversity in tropical ecosystems. *Crit. Iss. Biodiv.* 44–60.
- Collinge, S.K., 1996. Ecological consequences of habitat fragmentation: implications for landscape architecture and planning. *Land. Urb. Plann.* 36, 59–77.
- Colwell, R.K., 2000. Estatistical estimation of species richness and shared species from samples (EstimateS). 6.0b1.
- Colwell, R.K., Coddington, J.A., 1994. Estimating terrestrial biodiversity through extrapolation. *Philos. Trans. R. Soc. Lond.* 345, 101–118.
- Conrad, K.F., Pritchard, G., 1992. An ecological classification of odonate mating systems: the relative influence of natural, inter- and intra-sexual selection on males. *Biol. J. Linn. Soc.* 45, 255–269.
- Copeland, R.S., Okeka, W., Corbet, P.S., 1996. Treeholes as larval habitat of the dragonfly *Hemithysa camarensis* (Odonata: Libellulidae) in Kakamega forest, Kenya. *Aquat. Ins.* 18, 129–147.
- Corbet, P.S., 1980. Biology of Odonata. *Annu. Rev. Ent.* 25, 189–217.
- Corbet, P.S., 1999. Dragonflies: Behavior and Ecology of Odonata. Comstock Publ. Assoc., Ithaca, NY.
- Corbet, P.S., May, M.L., 2008. Fliers and perchers among Odonata: dichotomy or multidimensional continuum? A provisional reappraisal. *Int. J. Odonatol.* 11, 155–171.

- Corbi, J.J., Trivinho-Strixino, S., dos Santos, A., 2008. Environmental evaluation of metals in sediments and dragonflies due to sugar cane cultivation in neotropical streams. *Water Air Soil Pollut.* 195, 325–333.
- Costa, J.M., Lourenço, A.N., Vieira, L.P., 2002. *Micrathyrina pseudohypodidyma* sp. n. (Odonata: Libellulidae), com chave das espécies do gênero que ocorrem no Rio de Janeiro. *Neotrop. Entomol.* 31, 377–389.
- De Marco Jr., P., 1998. The Amazonian Campina dragonfly assemblage: patterns in microhabitat use and behavior in a foraging habitat. *Odonatologica* 27, 239–248.
- De Marco Jr., P., Coelho, F.M., 2004. Services performed by the ecosystem: forest remnants influence agricultural cultures' pollination and production. *Biodivers. Conserv.* 13, 1245–1255.
- De Marco Jr., P., Latini, A.O., Resende, D.C., 2005. Thermoregulatory constraints on behavior: patterns in a neotropical dragonfly assemblage. *Neotrop. Entomol.* 34, 155–162.
- De Marco Jr., P., Peixoto, P.E.C., 2004. Population dynamics of *Hetaerina rosea* and its relationship to abiotic conditions (Zygoptera: Calopterygidae). *Odonatologica* 33, 17–25.
- De Marco Jr., P., Resende, D.C., 2002. Activity patterns and thermoregulation in a tropical dragonfly assemblage. *Odonatologica* 31, 129–138.
- De Marco Jr., P., Vital, M.V.C., 2008. Ecology of *Tigriagrion aurantinigrum* calvert in response to variations in environmental conditions (Zygoptera: Coenagrionidae). *Odonatologica* 37, 1–11.
- Death, R.G., 1996. The effect of habitat stability on benthic invertebrate communities: the utility of species abundance distributions. *Hydrobiologia* 317, 97–107.
- Delacerda, L.D., Salomons, W., Pfeiffer, W.C., Bastos, W.R., 1991. Mercury distribution in sediment profiles from lakes of the high pantanal, Mato Grosso State, Brazil. *Biogeochemistry* 14, 91–97.
- ESRI, 1999. Arc View GIS 3.2a. Environmental Systems Research Institute, Inc., Redlands, CA.
- Ferreira-Peruquetti, P., De Marco Jr., P., 2002. Efeito da alteração ambiental sobre comunidades de Odonata em riachos de Mata Atlântica de Minas Gerais, Brasil. *Rev. Bras. Zool.* 19, 317–327.
- Ferreira-Peruquetti, P., Fonseca-Gessner, A.A., 2003. Comunidade de Odonata (Insecta) em áreas naturais de Cerrado e Monocultura no nordeste do Estado de São Paulo, Brasil: relação entre o uso do solo e a riqueza faunística. *Rev. Bras. Zool.* 20, 219–224.
- Fincke, O.M., 1984. Giant damselflies in a tropical forest: reproductive biology of *Megaloprepus coerulatus* with notes on *Mecistogaster* (Zygoptera: Pseudostigmatidae). *Adv. Odonat.* 2, 13–27.
- Foot, A.L., Hornung, C.L.R., 2005. Odonates as biological indicators of grazing effects on Canadian prairie wetlands. *Ecol. Entomol.* 30, 273–283.
- Golfari, L., 1975. Zoneamento ecológico do Estado de Minas para reflorestamento. CPRFC, Belo Horizonte, MG.
- Gray, J.S., 1981. Detecting pollution induced changes in communities using the log-normal distribution of individuals among species. *Mar. Pollut. Bull.* 12, 173–176.
- Harvey, I.F., Corbet, P.S., 1985. Territorial behaviour of larvae enhances mating success of male dragonflies. *Anim. Behav.* 33, 561–565.
- Helshe, J.F., Forrester, N.E., 1983. Estimating species richness using the Jackknife procedure. *Biometrics* 39, 1–11.
- Hill, J.K., Hamer, K.C., 1998. Using species abundance models as indicators of habitat disturbance in tropical forests. *J. Appl. Ecol.* 35, 458–460.
- Hillebrand, H., Blenckner, T., 2002. Regional and local impact on species diversity—from pattern to processes. *Oecologia* 132, 479–491.
- Jacobs, M.E., 1955. Studies on territorialism and sexual selection in dragonflies. *Ecology* 36, 566–586.
- Johansson, F., 1991. Foraging modes in an assemblage of odonate larvae: effects of prey and interference. *Hydrobiologia* 209, 79–87.
- Juen, L., Cabette, H.S.R., de Marco, P., 2007. Odonate assemblage structure in relation to basin and aquatic habitat structure in Pantanal wetlands. *Hydrobiologia* 579, 125–134.
- Kalcounis-Rueppell, M.C., Payne, V.H., Huff, S.R., Boyko, A.L., 2007. Effects of wastewater treatment plant effluent on bat foraging ecology in an urban stream system. *Biol. Conserv.* 138, 120–130.
- Karr, J.R., 1991. Biological integrity: a long-neglected aspect of water resource management. *Ecol. Appl.* 1, 66–84.
- Kay, J.J., 1993. On the nature of ecological integrity: some closing comments. In: Woodley, S., Kay, J., Francis, G.C. (Eds.), *Ecological Integrity and the Management of Ecosystems*. St. Lucie Press, FL, pp. 201–212.
- Kevan, P.G., Greco, C.F., Belousoff, S., 1997. Log-normality of biodiversity and abundance in diagnosis and measuring of ecosystemic health: pesticide stress on pollinators on blueberry heaths. *J. Appl. Ecol.* 34, 1122–1136.
- Kogan, M., Lattin, J.D., 1993. Insect conservation and pest management. *Biodivers. Conserv.* 2, 242–257.
- Lachavanne, J.B., Perfetta, J., Juge, R., 1992. Influence of water eutrophication on the macrophytic vegetation of Lake Lugano. *Aquat. Sci.* 54, 351–363.
- Lencioni, F.A.A., 2005. Damselflies of Brazil. An Illustrated Identification Guide. I. The Non-Coenagrionidae Families. All Print Editora, São Paulo, Brazil.
- Lencioni, F.A.A., 2006. Damselflies of Brazil. An Illustrated Identification Guide. II. Coenagrionidae Families. All Print Editora, São Paulo, Brazil.
- Leunda, P.M., Oscoz, J., Miranda, R., Arino, A.H., 2009. Longitudinal and seasonal variation of the benthic macroinvertebrate community and biotic indices in an undisturbed Pyrenean river. *Ecol. Indic.* 9, 52–63.
- Magurran, A.E., 1988. *Ecological Diversity and its Measurement*. Cambridge University Press, London.
- Mainstone, C.P., Parr, W., 2002. Phosphorus in rivers ecology and management. *Sci. Total Environ.* 282–283, 25–47.
- May, M.L., 1976. Thermoregulation in adaptation to temperature in dragonflies (Odonata: Anisoptera). *Ecol. Monogr.* 46, 1–32.
- May, M.L., 1977. Thermoregulation and reproductive activity in tropical dragonflies of the genus *Micrathyrina*. *Ecology* 58, 787–798.
- May, M.L., 1991. Thermal adaptations of dragonflies, revisited. *Adv. Odonat.* 5, 71–88.
- May, M.L., 1998. *Macrothemis fallax*, a new species of dragonfly from Central America (Anisoptera: Libellulidae), with a key to male *Macrothemis*. *Int. J. Odonatol.* 1, 137–153.
- May, R.M., 1975. Patterns of species abundance and diversity. In: Cody, M.L., Diamond, J.M.C.P. (Eds.), *Ecology and Evolution of Communities*. Harvard University Press, VL, pp. 81–120.
- May, R.M., 1982. Patterns of species abundance and diversity. In: Cody, M.L., Diamond, J.M.C.P. (Eds.), *Ecology and Evolution of Communities*. The Belknap Press of Harvard University Press, VL, pp. 81–120.
- McGill, B.J., Etienne, R.S., Gray, J.S., Alonso, D., Anderson, M.J., Benetcha, H.K., Dornelas, M., Enquist, B.J., Green, J.L., He, F.L., Hurlbert, A.H., Magurran, A.E., Marquet, P.A., Maurer, B.A., Ostling, A., Soykan, C.U., Ugland, K.I., White, E.P., 2007. Species abundance distributions: moving beyond single prediction theories to integration within an ecological framework. *Ecol. Lett.* 10, 995–1015.
- McPeck, M.A., 1989. Differential dispersal tendencies among *Enallagma damselflies* (Odonata) inhabiting different habitats. *Oikos* 56, 187–195.
- Michiels, N.K., Dhondt, A.A., 1989. Differences in male and female activity patterns in the dragonfly *Sympetrum danae* (Sulzer) and their relation to mate-finding (Anisoptera: Libellulidae). *Odonatologica* 18, 349–364.
- Michiels, N.K., Dhondt, A.A., 1991. Sources of variation in male mating success and female oviposition rate in a nonterritorial dragonfly. *Behav. Ecol. Sociobiol.* 29, 17–26.
- Mottet, A., Ladet, S., Coque, N., Gibon, A., 2006. Agricultural land-use change and its drivers in mountain landscapes: a case study in the Pyrenees. *Agric. Ecosyst. Environ.* 114, 296–310.
- Muttamara, S., 1996. Wastewater characteristics. *Resour. Conserv. Recycl.* 16, 145–159.
- Naranjo, S.E., Ellsworth, P.C., Hagler, J.R., 2004. Conservation of natural enemies in cotton: role of insect growth regulators in management of *Bemisia tabaci*. *Biol. Control* 30, 52–72.
- Nelson, W.G., 1987. An evaluation of deviation from the lognormal distribution among species as a pollution indicator in marine benthic communities. *J. Exp. Mar. Biol. Ecol.* 113, 181–206.
- Nessimian, J.L., Venticinque, E., Zuanon, J., de Marco, P., Gordo, M., Fidelis, L., Batista, J.D., Juen, L., 2008. Land use, habitat integrity, and aquatic insect assemblages in Central Amazonian streams. *Hydrobiologia* 614, 117–131.
- Oertli, B., 2008. The use of dragonflies in the assessment and monitoring of aquatic habitats. In: Cordoba-Aguilar, A. (Ed.), *Model Organisms for Ecological and Evolutionary Research*. Oxford University Press, Oxford, pp. 79–95.
- Paoletti, M.G., 1999. Using bioindicators based on biodiversity to assess landscape sustainability. *Agric. Ecosyst. Environ.* 74, 1–18.
- Pinto-Coelho, R.M., 1998. Effects of eutrophication on seasonal patterns of mesoplankton in a tropical reservoir: a 4-year study in Pampulha lake, Brazil. *Fresh. Biol.* 40, 159–173.
- Preston, F.W., 1948. The commonness and rarity of species. *Ecology* 29, 254–283.
- Preston, F.W., 1962. The canonical distribution of commonness and rarity. *Ecology* 43, 185–215.
- Samways, M.J., Steytler, N.S., 1996. Dragonfly (Odonata) distribution patterns in urban and forest landscapes, and recommendations for riparian management. *Biol. Conserv.* 78, 279–288.
- Schindler, M., Fesil, C., Chovanec, A., 2003. Dragonfly associations (Insecta: Odonata) in relation to habitat variables: a multivariate approach. *Hydrobiologia* 497, 169–180.
- Schultz, H., Waringer, J.A., Chovanec, A., 2003. Assessment of the ecological status of Danubian floodplains at Tulln (Lower Austria) based on the Odonata Habitat Index (OHI). *Odonatologica* 32, 355–370.
- Schutte, G., Reich, M., Plachter, H., 1997. Mobility of the rheobiont damselfly *Calopteryx splendens* (Harris) in fragmented habitats (Zygoptera: Calopterygidae). *Odonatologica* 26, 317–327.
- Simiao-Ferreira, J., DeMarco, P., Mazao, G.R., Carvalho, A.R., 2009. Chironomidae assemblage structure in relation to organic enrichment of an aquatic environment. *Neotrop. Entomol.* 38, 464–471.
- Skalski, T., Pospiech, N., 2006. Beetles community structures under different reclamation practices. *Eur. J. Soil Biol.* 42, S316–S320.
- Smith, J., Samways, M.J., Taylor, S., 2007. Assessing riparian quality using two complementary sets of bioindicators. *Biodivers. Conserv.* 16, 2695–2713.
- Smith, V.H., Tilman, G.D., Nekola, J.C., 1999. Eutrophication: impacts of excess nutrient inputs on freshwater, marine, and terrestrial ecosystems. *Environ. Pollut.* 100, 179–196.
- Solimani, A.G., Tarallo, G.A., Carchini, G., 1997. Life history and species composition of the damselfly assemblage along the urban tract of a river in central Italy. *Hydrobiologia* 356, 21–32.
- Steffan-Dewenter, I., Tscharntke, T., 2002. Insect communities and biotic interactions on fragmented calcareous grasslands—a mini review. *Biol. Conserv.* 104, 275–284.

- Suhling, F., Sahlen, G., Martens, A., Marais, E., Schutte, C., 2006. Dragonfly assemblages in arid tropical environments: a case study from Western Namibia. *Biodivers. Conserv.* 15, 311–332.
- Summerville, K.S., Crist, T.O., 2001. Effects of experimental habitat fragmentation on patch use by butterflies and skippers (Lepidoptera). *Ecology* 82, 1360–1370.
- Taylor, P.D., Merriam, G., 1995. Wing morphology of a forest damselfly is related to landscape structure. *Oikos* 73, 43–48.
- Tsirtsis, G., Spatharis, S., Karydis, M., 2008. Application of the lognormal equation to assess phytoplankton community structural changes induced by marine eutrophication. *Hydrobiologia* 605, 89–98.
- Utz, R.M., Hilderbrand, R.H., Boward, D.M., 2009. Identifying regional differences in threshold responses of aquatic invertebrates to land cover gradients. *Ecol. Indic.* 9, 556–567.
- Valverde, O., 1958. Estudo regional da Zona da Mata de Minas Gerais. *R. Bras. Geogr.* 20, 3–79.
- Watson, J.A.L., Arthington, A.H., Conrick, D.L., 1982. Effect of sewage effluent on dragonflies (Odonata) of Bulimba creek, Brisbane. *Aust. J. Mar. Fresh. Res.* 23, 517–528.
- Westfall Jr., M.J., 1984. Odonata. In: Merritt, R.W., Cummins, K.W.C.P. (Eds.), *An Introduction to the Aquatic Insects of North America* Kendall/Hunt Publishing Company, pp. 126–176.
- Wolf, L.L., Waltz, E.C., 1988. Oviposition site selection and spatial predictability of female white-faced dragonflies (*Leucorrhinia intacta*) (Odonata: Libellulidae). *Ethology* 78, 306–320.
- Zar, J.H., 1999. *Biostatistical Analysis*. Prentice-Hall, Englewood Cliffs, NJ.